

Healthcare Business Capstone – Comprehensive Grading Rubric (100%)

Phase 1 – SQL Analytics & Data Integrity (25%)

Criteria	Points	Description
Database Setup & Loading	8	CSV files loaded into SQL; correct schema, data types, and constraints applied.
Business Query Accuracy	9	Operational, financial, and data quality queries written correctly and optimized.
Business Insight Reporting	8	Clear, concise interpretation of SQL results in business terms.

Phase 2 – Exploratory Data Analysis & Feature Engineering (25%)

Criteria	Points	Description
Data Profiling & Missing Value Analysis	6	Missing values, invalid entries, and anomalies identified and handled appropriately.
Outlier Detection & Distribution Analysis	6	Distributions and outliers analyzed across departments, insurance providers, and cities.
Feature Engineering Quality	7	Patient, visit, and insurance-level features created with clear business justification.
Data Quality Documentation	6	Clean, well-structured data quality report with findings and actions.

Phase 3 – Classification Models & Training (25%)

Criteria	Points	Description
Baseline Model Implementation	6	Baseline models for both risk and claim classification trained and documented.
Advanced Models & Tuning	8	Tree-based or ensemble models implemented and hyperparameters tuned.
Train/Test Strategy & Leakage Control	6	Time-based split used and feature leakage prevented.

Model Artifact Management	5	Models and feature schema saved and versioned correctly.
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Phase 4 – Evaluation, Explainability & Governance (25%)

Criteria	Points	Description
Technical Evaluation Metrics	7	Confusion matrix, precision, recall, and F1-score computed for train and test sets.
Business Impact Metrics	6	High-risk recall and rejected-claim recall mapped to operational and financial impact.
Explainability & Feature Interpretation	6	Feature importance and model drivers explained in business language.
Fairness & Model Card	6	Performance segmented by gender, city, and insurance; model card documented.

Final Score Calculation

Total Score = (Phase 1 + Phase 2 + Phase 3 + Phase 4) / 100

Performance Evaluation

Score Range	Performance Level	Description
90–100%	Excellent	Professional-quality capstone with strong business and technical insights.
75–89%	Good	Solid implementation with minor technical or documentation gaps.
60–74%	Satisfactory	Basic implementation with limited depth of analysis or business impact.
Below 60%	Needs Improvement	Major gaps in execution, incomplete phases, or weak justification.

Final Capstone Deliverables

1. Jupyter Notebooks (Single OR Phase-wise)

Files:

- Healthcare_Capstone.ipynb

OR

- Phase1_SQL.ipynb, Phase2_EDA.ipynb, Phase3_Modeling.ipynb, Phase4_Evaluation.ipynb

Must include:

- Clean, commented code
- Plots and evaluation metrics
- Short business insights written in Markdown

2. API & Deployment Package

Includes:

- FastAPI source code
- Dockerfile
- Deployment runbook
- Sample API request and response documentation

3. Exported Data Outputs

Folder: data_outputs/

Includes:

- model_table.csv
- feature_schema.json
- drift_summary.csv (if implemented)

4. Final Business Report

File:

Healthcare_Insights_Report.docx (2–3 pages)

Contains:

- Key operational and financial findings
- Model impact summary
- Business recommendations for hospital leadership